

Tutorial 3 (Divide and Conquer Algorithms)

1. Explain the divide and conquer paradigm for algorithm design with suitable example.
2. Write down the binary search algorithm and analyze its complexity.
3. Trace the binary search algorithm for the following data and search key:
A = {-5, 1, 8, 24, 63, 84, 95, 99, 100}, Search key = 99
4. Trace the binary search algorithm for the following data and search key:
A = {1, 2, 3, 5, 7, 13, 17, 29, 53, 61, 85, 124}, Search key = 2
5. Write down the min-max algorithm and analyze its complexity.
6. Trace the min-max algorithm for the following data.
A = {50, 40, -5, -9, 45, 90, 65, 25, 75}
7. Trace the min-max algorithm for the following data.
A = {5, 7, 3, 4, 9, 12, 6, 2}
8. Trace the min-max algorithm for the following data.
A = {6, 4, 2, 3, 9, 8, 1, -8}
9. Write down the algorithm for merge operation and analyze its time and space complexities.
10. Write down the algorithm for merge sort and analyze its time and space complexity.
11. Trace the merge sort algorithm for the following data:
A = {9, 3, 17, 11, 6, 2, 1, 10}
12. Trace the merge sort algorithm for the following data:
A = {85, 76, 46, 92, 30, 41, 12}
13. Trace the merge sort algorithm for the following data:
A = {5, 2, 4, 6, 1, 3, 2, 6}
14. Discuss heapify operation with example. Write down its algorithm and analyze its time and space complexity.
15. Discuss build-max-heap operation with suitable example. Write down its algorithm and analyze its time and space complexity.
16. Write down the algorithm for heap sort and analyze its time complexity.
17. Trace the heap-sort algorithm for the following data:
A = {12, 45, 62, 50, 85, 15, 28}
18. Trace the heap-sort algorithm for the following data:
A = {16, 41, 18, 99, 74, 20, 17, 25, 10}
19. Trace the heap sort algorithm for the following data:
A = {4, 6, 17, 10, 12, 9, 8}
20. Write down the algorithm for quick sort and analyze its time complexity. Explain its best case and worst case complexities.
21. Trace the quick sort algorithm for the following data:
A = {15, 7, 6, 23, 18, 34, 25}
22. Trace the quick sort algorithm for the following data:
A = {99, 50, 60, 8, 5, 6, 20, 25, 40}
23. Write the Quick sort algorithm using randomized approach and explain its time complexity. How does this approach avoid the worst case?
24. Define order statistics.
25. Explain the quick select algorithm to make the selection of i^{th} smallest number in the given list in the expected linear time.
26. Explain the concept of selection in worst-case linear time.
27. Compare the time complexities of different sorting algorithms. (including those studied in chapter 2).